Why iid intervals fail with center correlation

The estimand is stable, but the uncertainty model changes.

Synthetic benchmark evidence boundary: generated DGP only, not completed validation or clinical evidence.

DGP

Y_ij = beta_0 + beta_1 T_ij + u_j + epsilon_ij

Estimand: beta_1 = treatment effect

u_j ~ N(0, tau^2)
center-level random effect
shared inside center j

epsilon_ij ~ N(0, sigma^2)
individual residual noise

ICC rho = tau^2 / (tau^2 + sigma^2)
center is the correlation and inference unit

Why this page matters: point estimates can remain centered near beta_1 while interval coverage fails when residuals are correlated within center.

Simulation grid: G=[8, 20, 50], rho=[0.0, 0.1, 0.3, 0.5], imbalance=['balanced', 'imbalanced'], seed=20260822.

Reference retrieval: statistical-model query selected inspected pages RRL-044, RRL-007, RRL-008, RRL-043, RRL-042; trace in EVIDENCE_MANIFEST; style not copied.

Cluster sandwich changes the variance object

The center, not the row, becomes the covariance aggregation unit.

Naive iid variance treats rows as independent; cluster sandwich groups score contributions by center.

Naive iid OLS interval
Var(beta_hat) = sigma_hat^2 (X'X)^(-1)
row residuals are pooled one by one

$$V_{CR} = (X'X)^{-1} \left[\sum_g X_g' u_g u_g' X_g \right] (X'X)^{-1}$$

Center g block
X_g rows + residual vector u_g aggregated before variance

Key change: the middle term preserves within-center covariance in the score. The current benchmark uses cluster-robust z intervals; small-G correction is reserved for the planned next experiment.

Synthetic method labels: naive_iid_ols_z versus cluster_robust_z. Both estimate the same beta_1; only the interval variance changes.

Reference retrieval: estimator/variance query selected inspected pages RRL-040, RRL-009, RRL-028, RRL-030, RRL-015; trace in EVIDENCE_MANIFEST; style not copied.

Simulation design tests interval behavior

DGP knobs flow into two interval methods and one endpoint gate.

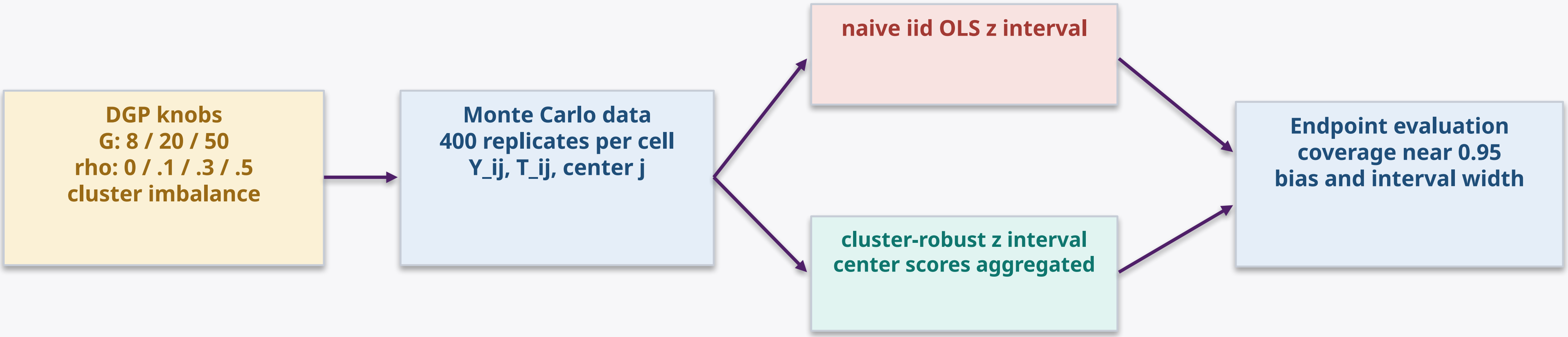


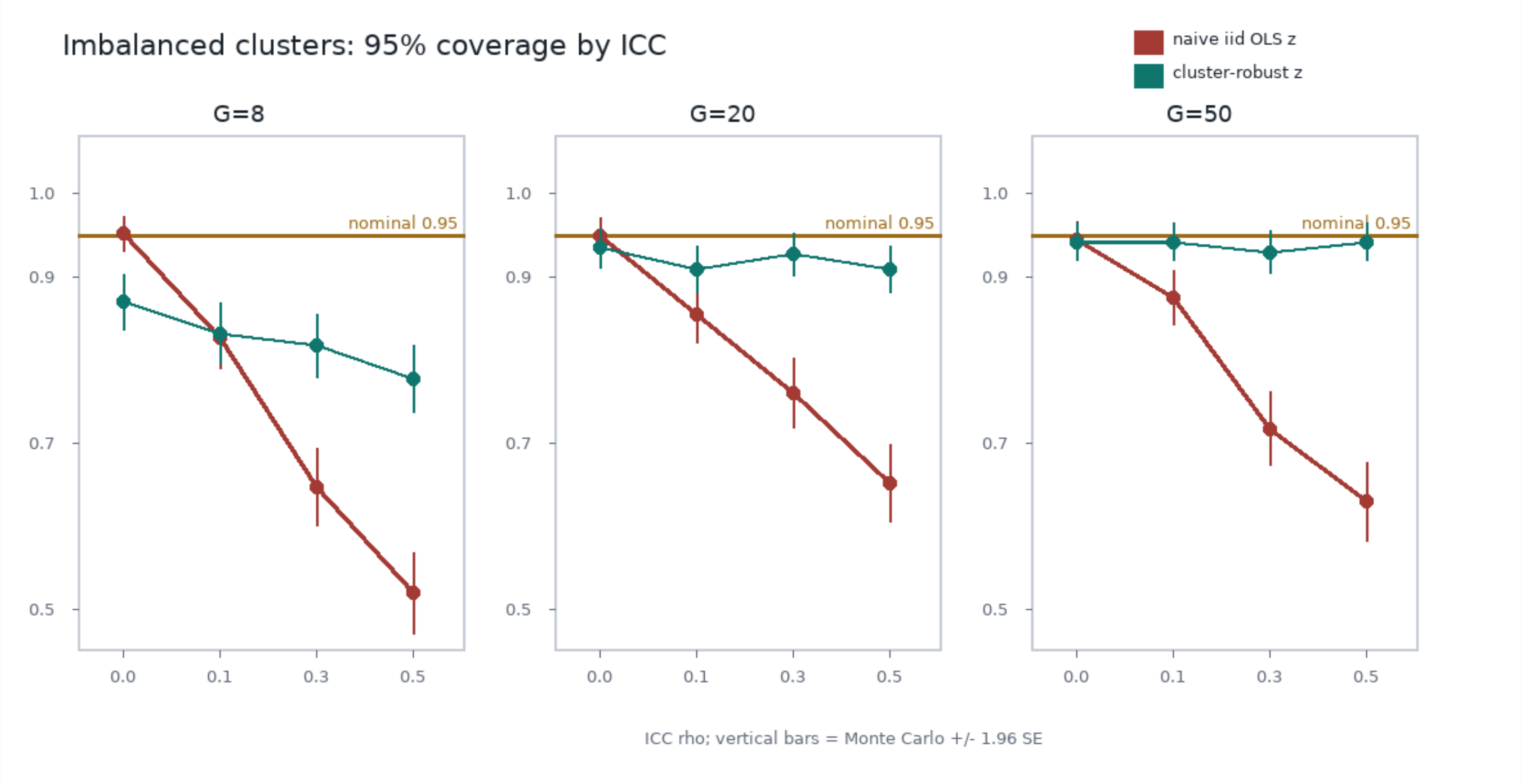
Diagram contract: structural connectors, visible arrowheads, left-to-right reading, no edge crossing.

Synthetic simulation evidence boundary: this experiment distinguishes interval behavior; it is not a general theorem.

Reference retrieval: simulation-design query selected inspected pages RRL-009, RRL-010, RRL-019, RRL-026, RRL-002; trace in EVIDENCE_MANIFEST; style not copied.

Coverage falls when ICC and imbalance rise

Coverage near 0.95 is the target; uncertainty must be visible.



Reading target
Coverage close to nominal 0.95 is the goal.
Above or below nominal is not automatically better.

Observed in this synthetic run
G=8, rho=.5, imbalanced:
naive=0.52
cluster=0.78

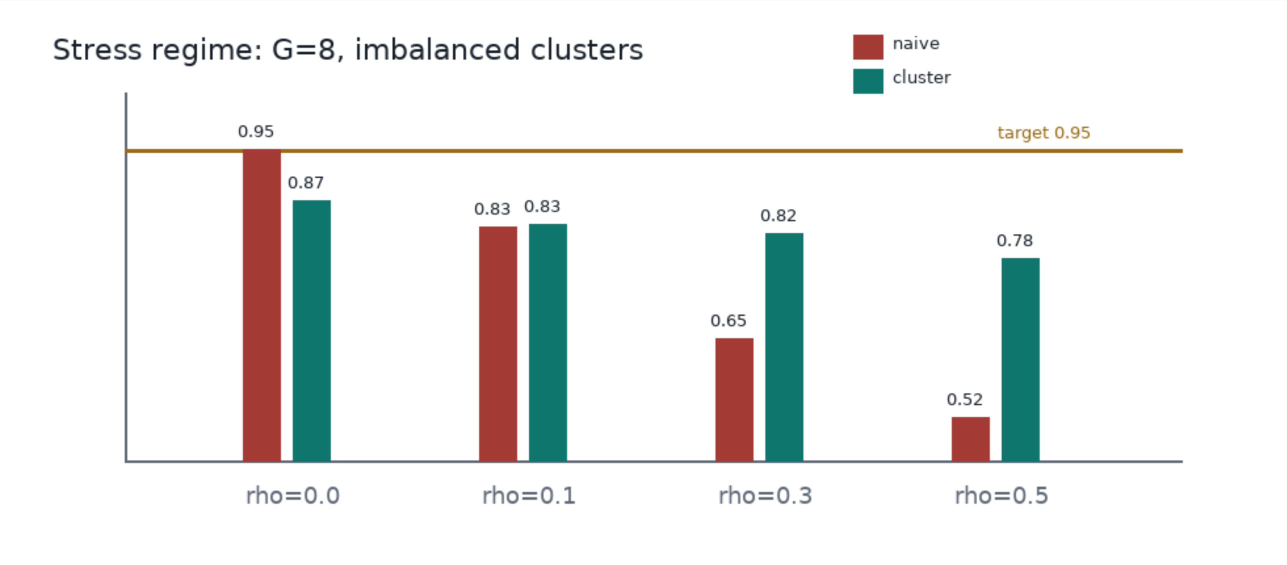
Conclusion limited to this simulation: cluster-robust intervals repair much of the iid undercoverage, but finite-center stress remains visible.

Synthetic simulation evidence boundary; error bars are Monte Carlo uncertainty from deterministic simulation output.

Reference retrieval: coverage-result query selected inspected pages RRL-030, RRL-033, RRL-010, RRL-023, RRL-039; trace in EVIDENCE_MANIFEST; style not copied.

Small-G stress remains the negative result

The next experiment must test finite-center corrections.



Negative result
G=8, rho=0.5, imbalanced cluster sizes and treatment shares
cluster-robust z coverage = 0.78

Failure mechanism
Few independent centers make the sandwich variance noisy; imbalance amplifies center-level leverage.

Next experiment is planned, not completed
Compare CR2 small-sample correction and wild cluster bootstrap on the same grid.

This page uses completed synthetic evidence only for the failure diagnosis; the proposed correction methods are explicitly future work.

Reference retrieval: negative-result query selected inspected pages RRL-011, RRL-024, RRL-025, RRL-041, RRL-005; trace in EVIDENCE_MANIFEST; style not copied.